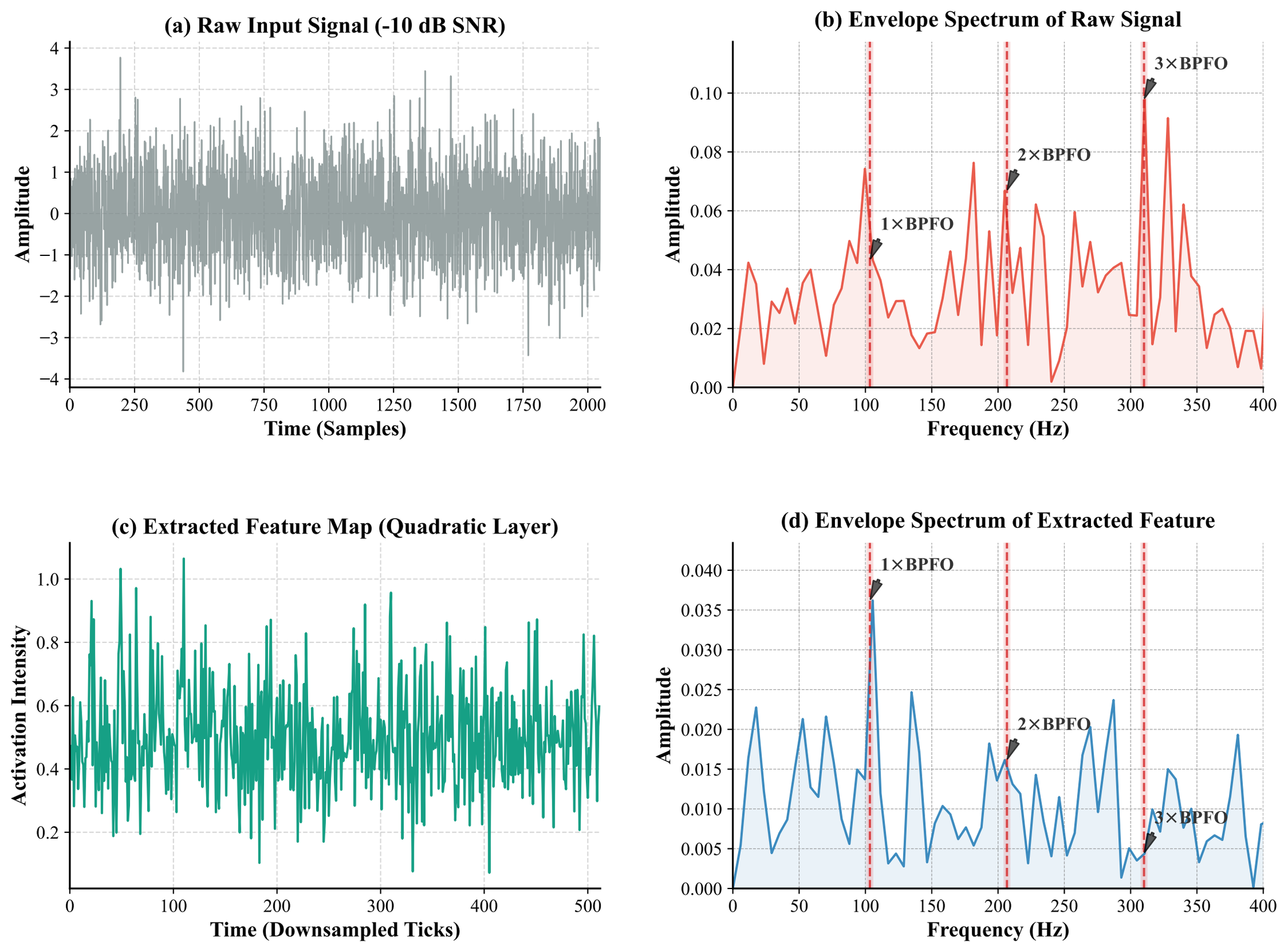

(a) Raw Input Signal (-10 dB SNR)
Amplitude
Time (Samples)
(b) Envelope Spectrum of Raw Signal
1×BPFO
2×BPFO
3×BPFO
Amplitude
Frequency (Hz)
(c) Extracted Feature Map (Quadratic Layer)
Activation Intensity
Time (Downsampled Ticks)
(d) Envelope Spectrum of Extracted Feature
1×BPFO
2×BPFO
3×BPFO
Amplitude
Frequency (Hz)